

- [10] P.-A. Moreau, F. Devaux, and E. Lantz, Einstein-Podolsky-Rosen paradox in twin images, *Phys. Rev. Lett.* **113**, 160401 (2014).
- [11] J. Schneeloch and G. A. Howland, Quantifying high-dimensional entanglement with Einstein-Podolsky-Rosen correlations, *Phys. Rev. A* **97**, 042338 (2018).
- [12] E. Giese, R. Fickler, W. Zhang, L. Chen, and R. W. Boyd, Influence of pump coherence on the quantum properties of spontaneous parametric down-conversion, *Phys. Scr.* **93**, 084001 (2018).
- [13] L. Chen, T. Ma, X. Qiu, D. Zhang, W. Zhang, and R. W. Boyd, Realization of the Einstein-Podolsky-Rosen paradox using radial position and radial momentum variables, *Phys. Rev. Lett.* **123**, 060403 (2019).
- [14] B. Ndagano, H. Defienne, A. Lyons, I. Starshynov, F. Villa, S. Tisa, and D. Faccio, Imaging and certifying high-dimensional entanglement with a single-photon avalanche diode camera, *npj Quantum Inf* **6**, 94 (2020).
- [15] V. Srivastav, N. H. Valencia, S. Leedumrongwatthanakun, W. McCutcheon, and M. Malik, Characterizing and tailoring spatial correlations in multimode parametric down-conversion, *Phys. Rev. Appl.* **18**, 054006 (2022).
- [16] A. Bhattacharjee, N. Meher, and A. K. Jha, Measurement of two-photon position-momentum Einstein-Podolsky-Rosen correlations through single-photon intensity measurements, *New J. Phys.* **24**, 053033 (2022).
- [17] A. Bhattacharjee, M. K. Joshi, S. Karan, J. Leach, and A. K. Jha, Propagation-induced revival of entanglement in the angle-oam bases, *Sci. Adv.* **8**, eabn7876 (2022).
- [18] S. Patil, S. Prabhakar, A. Biswas, A. Kumar, and R. P. Singh, Anisotropic spatial entanglement, *Phys. Lett. A* **457**, 128583 (2023).
- [19] A. G. da Costa Moura, W. A. T. Nogueira, and C. H. Monken, Fourth-order image formation by spontaneous parametric down-conversion: The effect of anisotropy, *Opt. Commun.* **283**, 2866 (2010).
- [20] E. S. Gómez, W. A. T. Nogueira, C. H. Monken, and G. Lima, Quantifying the non-gaussianity of the state of spatially correlated down-converted photons, *Opt. Express* **20**, 3753 (2012).
- [21] A. Yariv and P. Yeh, *Optical Waves in Crystals: Propagation and Control of Laser Radiation* (Wiley, 2002).
- [22] S. P. Walborn, C. H. Monken, S. Pádua, and P. H. Souto Ribeiro, Spatial correlations in parametric down-conversion, *Phys. Rep.* **495**, 87 (2010).
- [23] J. W. Goodman, *Introduction to Fourier Optics*, 3rd ed. (Roberts & Company, 2005).
- [24] A. E. Siegman, *Lasers* (University Science Books, 1986).
- [25] N. N. Lebedev, *Special Functions & Their Applications* (Dover, 1972).

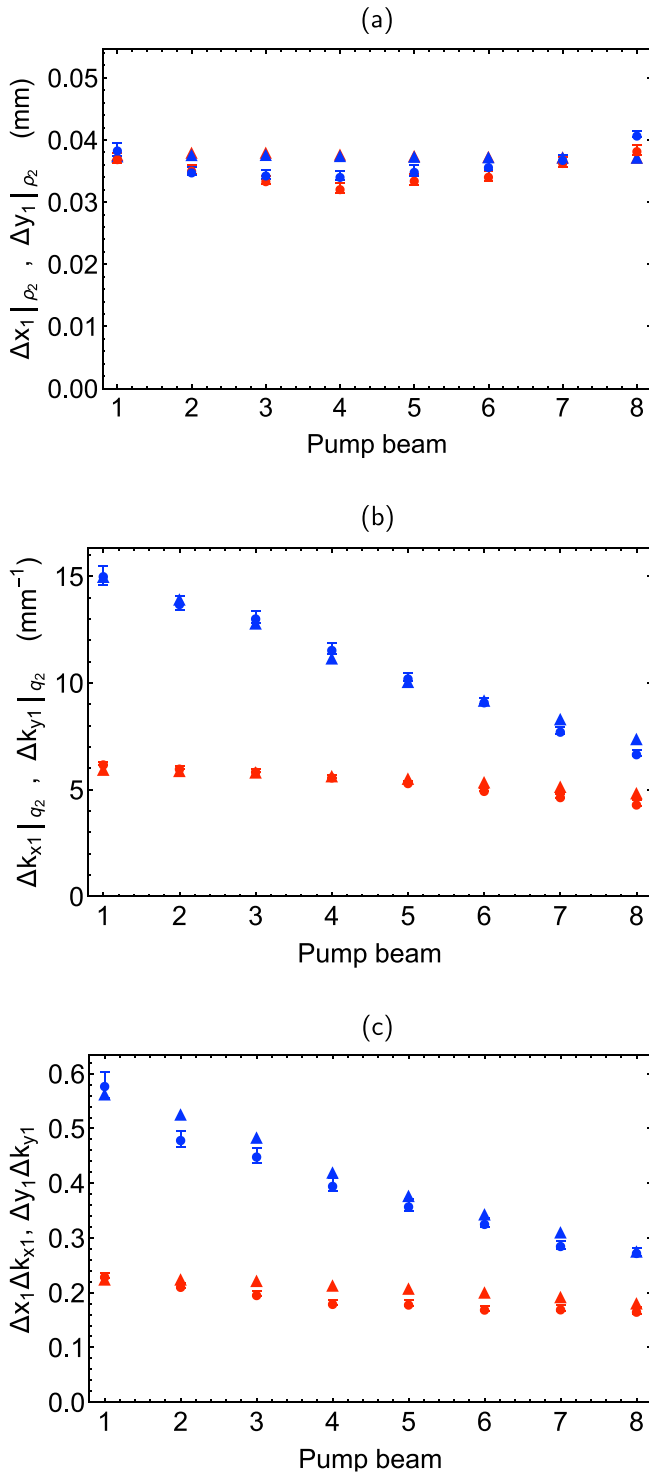


FIG. 7. Measured (●) and predicted (▲) values of (a) Δx_1 (red) and Δy_1 (blue), (b) Δk_{x1} (red) and Δk_{y1} (blue), (c) $\Delta x_1\Delta k_{x1}$ (red) and $\Delta y_1\Delta k_{y1}$ (blue) for a 5 mm-long BBO crystal pumped by 355 nm laser beams whose parameters are listed in Table I.